GIRIDHARAN S

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Summary

Computer Science undergraduate at VIT Vellore (CGPA 9.19) who builds intelligent software and robotics systems: ROS 2 mapping and navigation, neuro-symbolic AI agents, multi-objective optimization, explainable databases and compilers. Seeking internships in robotics, AI/ML or software engineering.

Skills

Languages: Python, C, C++, Java, JavaScript, SQL
AI/ML & Vision: scikit-learn, OpenCV, neuro-symbolic agents, model evaluation
Robotics: ROS 2, RViz, Gazebo, LiDAR, URDF, Nav2 (learning)
Frontend: React, Vite, HTML, CSS
Backend & Databases: FastAPI, Flask, PostgreSQL (PL/pgSQL, triggers), SQLite
Tools: Git, GitHub, VS Code, Jupyter Notebook, pytest, Mininet, ESP32
Concepts: Data Structures & Algorithms, OOP, DBMS, Compiler Design, Computer Vision

Experience

Junior Software Engineer **Jul 2026 – Present**
Team Artemis, VIT Vellore
- Mapped the Team Artemis lab (physical LiDAR) and a Gazebo maze in ROS 2 using Hexapod and TurtleBot platforms.
- Ran 2 navigation tests to predefined goals on saved maps; learning Nav2 and exploring a modified navigation approach.
- Supporting EYIC and Technocian finals preparation, and assisting juniors with RoboReach at Techfest, IIT Bombay.

AI/ML Intern **May 2025 – Jun 2025**
IITM Pravartak (IIT Madras), Chennai (Hybrid)
- Validated ∼800 Hindi speech-to-text test cases from Sansad TV broadcasts against ground-truth transcripts.
- Evaluated model outputs for accuracy, consistency and edge-case handling, and logged findings for model iteration.

Projects

Knowledge-Augmented Neuro-Symbolic Travel Planner | Python, Flask, SQLite, scikit-learn, pytest | GitHub
- Built an AI agent that turns natural-language trip requests into validated, constrained flight × hotel plans.
- Split a parse-only language layer from a symbolic core for hard constraints, ranking, risk tiers and V1–V7 validation.
- Plan-aware elicitation asks only when an answer changes the plan: 6 vs. 9 first-trip questions at 100% oracle match.
- Added SQLite preference memory, a pairwise logistic-regression ranker with EVOI questioning, and 158 pytest tests.

GreenPlace – SDN Controller Placement (Group Project) | Python, ILP, NSGA-II, Mininet, Ryu | GitHub
- Modelled energy-aware SDN controller placement under hard delay limits, controller power and migration burden.
- Solved static placement with an exact ILP baseline and NSGA-II across 46 real Internet Topology Zoo networks.
- Built a runtime manager for dynamic traffic; Mininet/Ryu emulation executed and verified 212 switch migrations.

CineTrace – Explainable and Adaptive Movie Recommender | React, FastAPI, PostgreSQL, pytest
- Designed an 18-table PostgreSQL schema linking every explanation by foreign key to the exact rows that justify it.
- Built trigger-maintained, append-only rating history and a time-travel SQL verifier: 0 of 4,294 claims invalid.
- Enforced integrity with CHECKs, deferred and immutability triggers; drift-aware item-kNN with MMR; 40 pytest tests.

Mini Compiler for a Custom Language (Team of 3) | Python, unittest, React
- Built a seven-phase compiler, from lexer and recursive-descent parser to a cycle-counting VM, with if/while and functions.
- Designed a cost-gated optimizer that keeps a rewrite only if cycles drop: 29.8% fewer VM cycles than standard passes.
- Added graph-colouring register allocation and a learned cost model; verified by 81 unit tests and differential testing.

Certifications & Achievements

- **Robotic Fundamentals and Introduction to ROS:** Value Added Course, VIT (Jan–Apr 2026), covering RViz, Gazebo and SLAM.
- **Machine Learning & AI Fundamentals:** Training program, TEACHNOOK, Remote (Oct–Dec 2024).
- **Problem Solving:** Solved 55+ problems on LeetCode (arrays, strings, dynamic programming, graphs).

Education

Vellore Institute of Technology (VIT), Vellore **Aug 2024 – May 2028**
B.Tech in Computer Science *CGPA: 9.19 / 10 (till date)*
Samsidh International School **2022 – 2024**
Class XII *85%*
Chinmaya Vidyalaya **2015 – 2022**
Class X *88%*